Ionic Bonding - Type II

Type I - Review

red + blue

- Type I - Binary compound formed from a metal (Group 1A, 2A, 3A, and Ag, Zn, or Cd) and a nonmetal
- Formulas
  - Write each ion, cation comes first.
  - Final formula must be neutral
  - If charges cancel, just write symbols
  - If charges do not cancel, use the criss-cross
  - Reduce subscripts if needed
- Names
  - Write the name of both ions, cation comes first
  - Change the ending of the anion to -ide

+1 Ø

THE PERIODIC TABLE OF THE ELEMENTS

CATIONS DO NOT NEED A ROMAN NUMERAL

+1 +2

Period	Group 1	+2 Group 2	Group 3	Group 4	Group 5	Group 6	Group 7	Group 8	Group 9	Group 10	+1 Group 11	+2 Group 12	+3 Group 13/3B	-4 Group 14/4B	-3 Group 15/5B	-2 Group 16/6B	-1 Group 17/7B	Ø Group 18/8B
Period 1	1 H Hydrogen 1.01																	2 He Helium 4.00
Period 2	3 Li Lithium 6.94	4 Be Beryllium 9.01											5 B Boron 10.81	6 C Carbon 12.01	7 N Nitrogen 14.01	8 O Oxygen 16.00	9 F Fluorine 19.00	10 Ne Neon 20.18
Period 3	11 Na Sodium 22.99	12 Mg Magnesium 24.30											13 Al Aluminum 26.98	14 Si Silicon 28.09	15 P Phosphorus 30.97	16 S Sulfur 32.07	17 Cl Chlorine 35.45	18 Ar Argon 39.95
Period 4	19 K Potassium 39.10	20 Ca Calcium 40.08	21 Sc Scandium 44.96	22 Ti Titanium 47.87	23 V Vanadium 50.94	24 Cr Chromium 52.00	25 Mn Manganese 54.94	26 Fe Iron 55.84	27 Co Cobalt 58.93	28 Ni Nickel 58.69	29 Cu Copper 63.55	30 Zn Zinc 65.39	31 Ga Gallium 69.72	32 Ge Germanium 72.61	33 As Arsenic 74.92	34 Se Selenium 78.96	35 Br Bromine 79.90	36 Kr Krypton 83.80
Period 5	37 Rb Rubidium 85.47	38 Sr Strontium 87.62	39 Y Yttrium 88.91	40 Zr Zirconium 91.22	41 Nb Niobium 92.91	42 Mo Molybdenum 95.94	43 Tc Technetium 98.00	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.91	46 Pd Palladium 106.42	47 Ag Silver 107.87	48 Cd Cadmium 112.41	49 In Indium 114.82	50 Sn Tin 118.71	51 Sb Antimony 121.76	52 Te Tellurium 127.60	53 I Iodine 126.90	54 Xe Xenon 131.29
Period 6	55 Cs Cesium 132.91	56 Ba Barium 137.33	71 Lu Lutetium 174.97	72 Hf Halfnium 178.49	73 Ta Tantalum 180.95	74 W Tungsten 183.84	75 Re Rhenium 186.21	76 Os Osmium 190.23	77 Ir Iridium 192.22	78 Pt Platinum 195.08	79 Au Gold 196.97	80 Hg Mercury 200.59	81 Tl Thallium 204.38	82 Pb Lead 207.20	83 Bi Bismuth 208.98	84 Po Polonium 209.00	85 At Astatine 210.00	86 Rn Radon 222.00
Period 7	87 Fr Francium 223.00	88 Ra Radium 226.00	103 Lr Lawrencium 262.00	104 Rf Rutherfordium 261.00	105 Db Dubnium 262.00	106 Sg Seaborgium 263.00	107 Bh Bohrium 264.00	108 Hs Hassium 269.00	109 Mt Meitnerium 266.00	110 Ds Darmstadtium 271.00	111 Rg Roentgenium 272.00	112 Cn Copernicium 277.00	113 Nh Nihonium 284.00	114 Fl Flerovium 289.00	115 Mc Moscovium 288.00	116 Lv Livermorium 293.00	117 Ts Tennessine 294.00	118 Og Oganesson 294.00

Lanthanide Series	57 La Lanthanum 138.91	58 Ce Cerium 140.12	59 Pr Praseodymium 140.91	60 Nd Neodymium 144.24	61 Pm Promethium 145.00	62 Sm Samarium 150.36	63 Eu Europium 151.96	64 Gd Gadolinium 200.59	65 Tb Terbium 158.93	66 Dy Dysprosium 162.5	67 Ho Holmium 164.93	68 Er Erbium 167.26	69 Tm Thulium 168.93	70 Yb Ytterbium 173.04
Actinide Series	89 Ac Actinium 227.00	90 Th Thorium 232.04	91 Pa Protactinium 231.04	92 U Uranium 238.03	93 Np Neptunium 237.00	94 Pu Plutonium 244.00	95 Am Americium 243.00	96 Cm Curium 247.00	97 Bk Berkelium 247.00	98 Cf Californium 251.00	99 Es Einsteinium 252.00	100 Fm Fermium 257.00	101 Md Mendelevium 258.00	102 No Nobelium 259.00

Examples

Magnesium oxide $Mg^{+2}O^{-2}$ MgO

Silver nitride $Ag^{+1}_{3}N^{-3}_{1}$ Ag_3N

Li_2S lithium sulfide

$AlBr_3$ aluminum bromide

Type II white + blue

- Type II - Binary compound formed from all metals except (1A, 2A, 3A, and Ag, Zn, or Cd) + a nonmetal
- Formulas
  - Write each ion, cation comes first.
  - Final formula must be neutral
  - If charges cancel, just write the symbols
  - If charges do not cancel, use the criss-cross
  - Reduce subscripts if needed

Type II

- Names
  - Write the names of both ions, cation comes first
  - Change the ending of anion to -ide
  - Stock System - You MUST use Roman Numerals to show the specific charge of the cations
  - The only metals you will not use a Roman Numeral for are: (Groups 1A, 2A, 3A, and Ag, Zn, and Cd).

THE PERIODIC TABLE OF THE ELEMENTS

CATIONS DO NOT NEED A ROMAN NUMERAL

	Group 1	Group 2	Group 3	Group 4	Group 5	Group 6	Group 7	Group 8	Group 9	Group 10	Group 11	Group 12	Group 13/3B	Group 14/4B	Group 15/5B	Group 16/6B	Group 17/7B	Group 18/8B
	+1										+1	+2	+3	-4	-3	-2	-1	Ø
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Old Names for some Type II Cations

Ions	Old System	Stock System
Fe^{+2}	Ferrous	Iron(II)
Fe^{+3}	Ferric	Iron(III)
Cu^{+1}	Cuprous	Copper(I)
Cu^{+2}	Cupric	Copper(II)
Sn^{+2}	Stannous	Tin(II)
Sn^{+4}	Stannic	Tin(IV)

Type II

Iron(III) sulfide $Fe^{+3}_{2}S^{-2}_{3}$ Fe_2S_3

manganese(IV) nitride $Mn^{+4}_{3}N^{-3}_{4}$ Mn_3N_4

Copper(II) oxide $Cu^{+2}O^{-2}$ CuO

lead(II) phosphide $Pb^{+2}_{3}P^{-3}_{2}$ Pb_3P_2

Type II

+3 −1
$CrCl_3$
1

chromium(III) chloride

+2 −3
Zn_3P_2

Zinc phosphide

+1 −2
Au_2O_1

gold(1) oxide

+3 −1
$PtBr_3$

platinum(III) bromide

(2)+2 PbO_2 -1(2) = -2

+4

lead(IV) oxide

$Pb^{+4}_2 O^{-2}_4$ Pb_2O_4

PbO_2

PbO_2

$x + 2(-2) = 0$

$x - 4 = 0$

$x = +4$

— alternative method

PbO_2